Certicom 130-bit challenge workflow integration

Recorded 2026-10-20; validation updated 2026-10-21 · Darwin arm64 · Rust/Cargo 1.93.1

Exact identities and command paths

The published Certicom curve list supplies the prime-field ECCp-131 group and the distinct binary Koblitz ECC2K-130 group. The ECCp-131 field, curve, generator, subgroup order, cofactor, and published point are bound in the curve constructor and IC profile. ECC2K-130 already has a fixed-parameter file in `aburan28/cryptanalysis`.

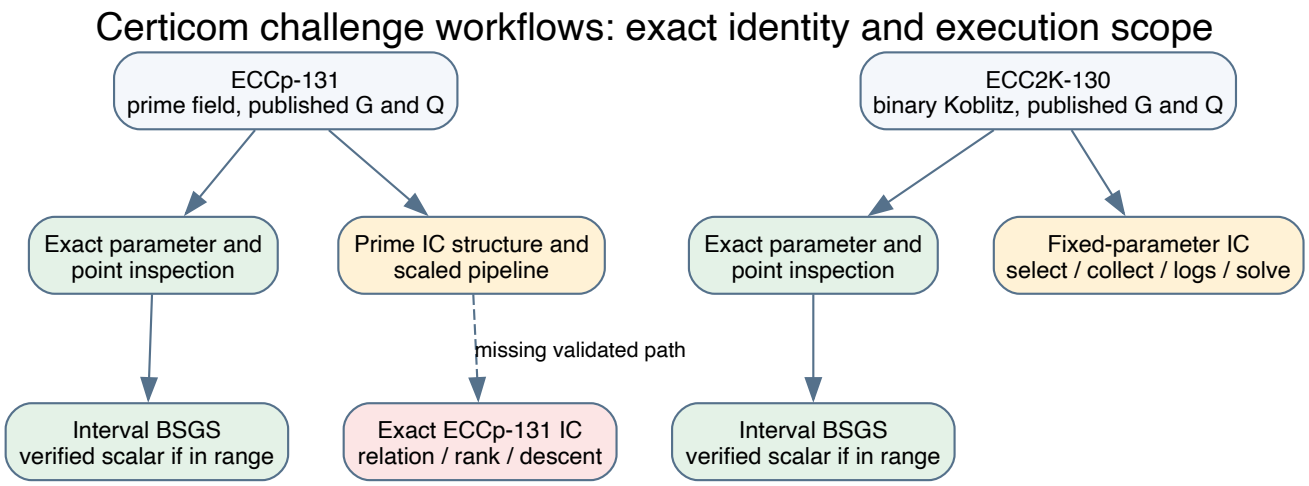


Figure 1: Exact-parameter checks and bounded BSGS are separate from the scaled prime IC and staged binary IC paths. The red box is an open implementation gap. Source: `workflow.dot`.

Observed validation

- `cryptanalysis` commit `3ac3f59363`: two ECCp-131 release unit tests passed; the direct `ca-ic inspect` command reported `checks_passed` for both published points, including curve-equation and subgroup checks. General prime-field point counting was explicitly `not_checked`.
- `ca-icx inspect eccp-131` passed; its estimate identified a scaled IC regime. `crax ecdlp --analyze-only` accepted the published point for structural analysis.
- The ECC2K-130 fixed-parameter status command exited 0 with `status: stopped`, zero saved relations, and no logarithm database. Its raw JSON and SQLite state are retained. The command was a workflow check; its operation counters are not used as an attack-cost comparison.
- `crypto` implementation commit `07dd1d97fa`: the exact-group bounded BSGS release test passed. The first full release library run ended with 4,445 passed, 11 failed, and 119 ignored. Seven failures came from a sparse-checkout fixture omission; all affected tests passed once its tracked files were present. Four unchanged time-limited `jv_cover` cases passed in isolation after reruns, with one intermediate sample-count failure. The failed full run remains recorded in the source report.
- The release `ecdlp-nist` CLI verified planted scalar 12345 on each exact group. Importing the published ECCp-131 point with a 64-bit BSGS interval exited with the stated 1,048,576-entry cap before group operations. The three raw JSON command outputs are retained with hashes.
- The release `ic inspect --curve eccp-131 --json` command reported `checks_passed` for both published points' curve-equation and subgroup checks. General prime-field point counting remained `not_checked`; the raw JSON is retained.

Boundary and next check

The exact ECCp-131 relation-to-scalar chain has no validated full-parameter run. The prime IC CLI executes its relations and descent on a generated smaller curve; the problem record states the required exact-curve relation, independent rank, target descent, reconstruction, and cold-cost checks. Bounded BSGS requires a justified interval and refuses a whole-subgroup table. No speed or full-parameter IC cost conclusion is made here.

The existing index-calculus scoreboard and progress timeline were checked. Neither has a new measured row from this integration, so their canonical data and rendered copies remain current. Full commands, revisions, initial sparse-checkout setup failures, raw artifact digests, and status distinctions are in REPORT.md.